

BATS

The only mammals capable of powered flight, bats are primarily nocturnal. Many species use echolocation to navigate and find food.

Bats are found worldwide in many different habitats, including tropical, subtropical, and temperate forests, savanna grasslands, deserts, and wetlands. Most fruit bats, as their name suggests, eat fruits. All other bats, once referred to as microbats, are mainly insect eaters. However, some bats drink nectar and eat pollen, a few suck blood, and some eat vertebrates, such as fish, frogs, and bats.

Their greatly elongated arm, hand, and finger bones support an elastic wing membrane for flight. Many bats also have a tail membrane between their legs. Bats typically rest upside down, hanging from their strong toes and claws.

THE SENSE OF ECHOLOCATION

Fruit bats rely predominantly on vision and smell, whereas microbats use a specialized sense called echolocation to avoid hitting objects, and to detect prey in the dark. They emit pulses of sound through their mouth or nose and form a “sound”

image of their surroundings from returning echoes. Those species that emit echolocation calls through their nose usually have elaborate facial ornamentation, called noseleaves, for focusing the sound. Bats have very sensitive hearing, often highly tuned to the frequency of their returning echoes. Some species listen to prey-generated sounds, such as the rustle that an insect makes when walking on a leaf.

HABITS AND ADAPTATIONS

Bats are very social animals, living in colonies of hundreds or thousands and, exceptionally, millions of animals. They roost in trees and caves or in buildings, bridges, and mines. Temperate species either migrate to warmer climates or hibernate during winter. They may also become torpid if food becomes short in other seasons. Many interesting reproductive adaptations have developed, including the storage of sperm, delayed fertilization, and delayed implantation to ensure that the young are born at the optimum time of year.

PHYLUM	CHORDATA
CLASS	MAMMALIA
ORDER	CHIROPTERA
FAMILIES	21
SPECIES	About 1,400

DEBATE EVOLUTIONARY DEBATE

Morphological and genetic analyses of the relationships between bat families do not always agree with one another. Genetic studies suggest that all bats evolved from one common ancestor and that flight evolved only once. However, molecular studies, although still proposing two bat groups, also indicate that some echolocating bats, such as the horseshoe bats, are more closely related to non-echolocating fruit bats (Yinpterochiroptera) than to other echolocating microbats (Yangochiroptera) — an apparent conundrum unless echolocation has evolved twice or has later been lost in the fruit bats.

FRUIT BATS

The bats of the family Pteropodidae are distributed across the tropical and subtropical regions of the Old World. They have a dog-like face with simple ears and large eyes. They use vision and smell to locate their food, except for members of the genus *Rousettus*, which echolocate by producing tongue clicks. These bats feed on fruits, nectar, and pollen. They have claws on both the thumb and the second finger.

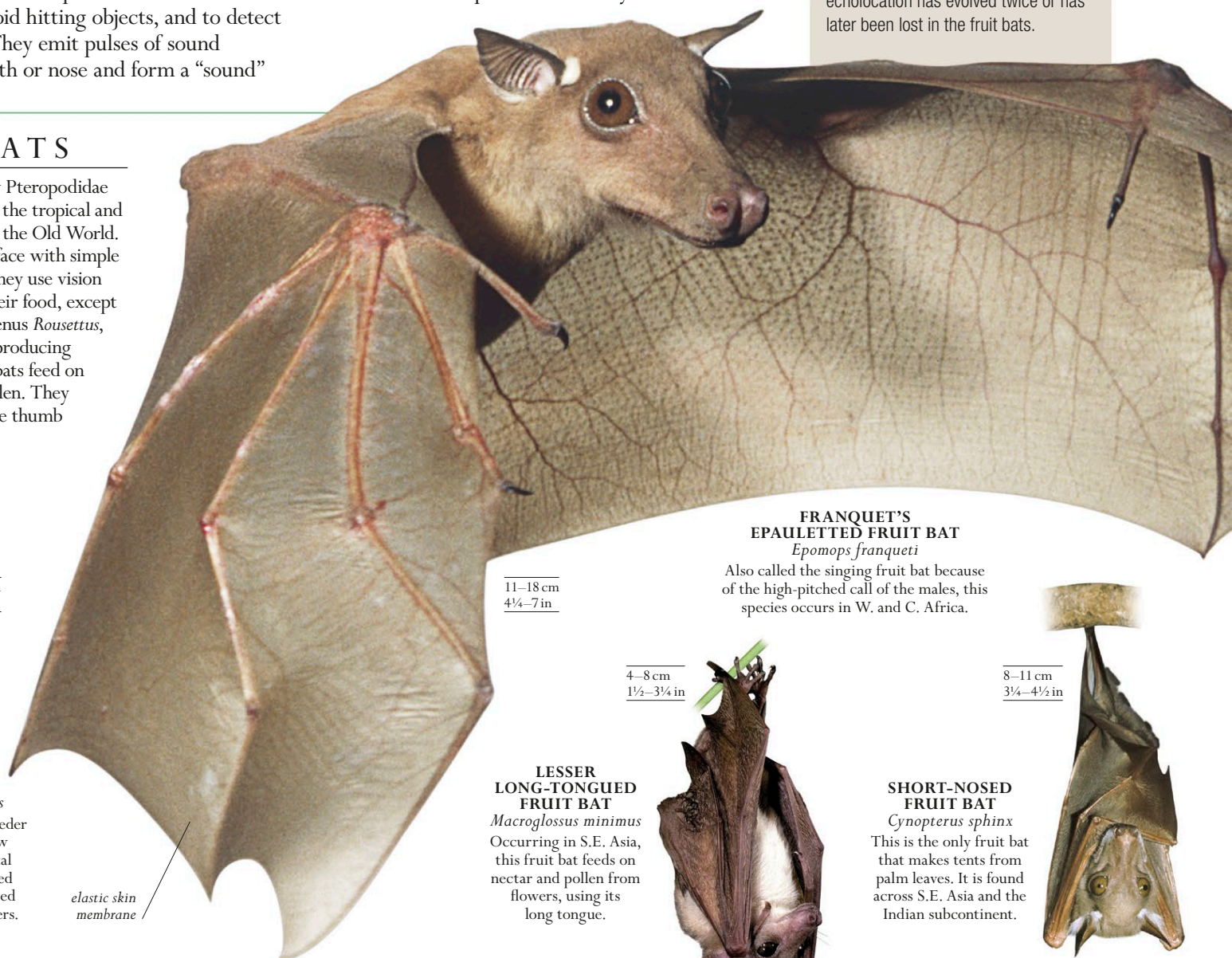


5–7.5 cm
2–3 in

BLOSSOM BAT
Syconycteris australis

This specialized nectar-feeder occurs from Papua New Guinea to eastern coastal Australia. It has a pointed muzzle and a brush-tipped tongue for probing flowers.

elastic skin
membrane



11–18 cm
4¼–7 in

LESSER LONG-TONGUED FRUIT BAT
Macroglossus minimus

Occurring in S.E. Asia, this fruit bat feeds on nectar and pollen from flowers, using its long tongue.

4–8 cm
1½–3¼ in



FRANQUET'S EPAULETTED FRUIT BAT
Epomops franqueti

Also called the singing fruit bat because of the high-pitched call of the males, this species occurs in W. and C. Africa.

8–11 cm
3¼–4½ in



SHORT-NOSED FRUIT BAT
Cynopterus sphinx

This is the only fruit bat that makes tents from palm leaves. It is found across S.E. Asia and the Indian subcontinent.